Vanja Smailovic - Portfolio

Lead AI Engineer and Applied AI Consultant

Services

I help product and engineering teams turn ambiguous AI opportunities into shipped product capabilities:

- LLM agents and AI assistants
- Document extraction, structured outputs and RAG
- Machine learning
- Text-to-SQL and analytics agents
- Anomaly detection and monitoring products
- Applied AI prototyping and MVP delivery
- AI product discovery, evaluation and technical roadmap



Case Studies



1. CEO-facing AI Executive Assistant

Problem	CEO office faced high-volume, high-context information workflows around email triage and daily briefings.
Built	Multi-agent AI assistant for email classification, briefing and source-grounded summaries.
Tech	Python, Google ADK, Gemini, Claude, Vertex AI, LiteLLM, Pydantic, Google and Slack APIs.

Outcome	Active initial rollout in ~6 weeks, estimated 2.5 hours saved per week in executive email reading time.



2. AI Invoice PDF Parser

Problem	Monthly invoice extraction was manual, slow and error-prone.
Built	Live, in-house parser converting invoice PDFs into validated structured data.
Tech	Python, OpenAI Structured Outputs, multimodal GPT-4.1, OCR, PyMuPDF, Pydantic.
Outcome	Reduced one FTE day of manual work to ~20 minutes, a 24x reduction, with ~99% typical accuracy.



3. Text-to-SQL AI Payments Analytics Agent

Problem	Customers needed self-serve payments analytics without SQL or domain knowledge.
Built	Customer-facing analytics agent translating natural language into grounded payments queries.
Tech	Python, GPT-4o, Chainlit, tool calling, schema grounding, SQL, FAISS, light RAG over company documentation.
Outcome	Enabled live natural-language access to expert-level payments insights and reduced reliance on manual analytics workflows.



4. Payments Anomaly Detection Product

Problem	Merchants needed monitoring and alerting for unusual payments behavior.
Built	Customer-facing anomaly detection product for merchant payments monitoring and alerts.
Tech	Python, SQL, Meta Prophet, Redshift, Cube.js, payments data transformation, querying optimization, anomaly detection, machine learning.
Outcome	Became the company's second most important product area, adopted by approximately 20 active B2B customers.



5. Credit Default Risk Modeling

Problem	Credit providers needed stronger risk assessment from raw bank transaction data.
Built	Credit risk capabilities using transaction tagging, NLP classification and binary default modeling.
Tech	Python, SQL, bank transaction data, XGBoost, BoostARoota, SHAP, AWS, Google Cloud.

Outcome	Supported revenue-generating credit products with more than 5 million credit decisions every year across 150 live customers and 13 markets, and supported technical delivery for a major customer acquisition.



6. Manufacturing Cost-Savings Regression Model

Problem	Inconsistent layer thickness affected quality in tungsten-carbide metal-cutting insert production.
Built	Regression analysis identifying key predictors of layer thickness from furnace sensor signals.
Tech	Python, pandas, regression modeling, statistical analysis, 50+ sensor signals, manufacturing process data.
Outcome	Supported quality-control improvements and identified estimated savings of €800k+.



7. Manufacturing Sales-Prioritization Map

Problem	Sales teams needed prioritized US machine-shop prospects for targeted outreach.
Built	Interactive geospatial map categorizing prospects by estimated commercial potential.
Tech	Python, web scraping, data cleaning, business prioritization logic, geospatial visualization.
Outcome	Helped sales teams navigate prospects geographically and focus outreach on higher-potential opportunities.



8. Bank Transaction Anonymization Pipeline

Problem	Strict regulatory requirements prevented data teams from accessing bank transaction data essential to daily analysis and product development.
Built	Compliance-approved anonymization pipeline preserving the analytical value of transaction data for safe, organization-wide use.
Tech	Python, SQL, Apache Airflow, AWS Athena, Amazon S3, AWS Glue, PySpark, pandas, Google Cloud Storage, Google BigQuery.
Outcome	Unlocked organization-wide access to previously inaccessible transaction data, creating a continuously growing resource for analysis and product innovation.



Clients

EQT

pagos

tink
A Visa Solution

instantor

SANDVIK

ERICSSON

Leadership



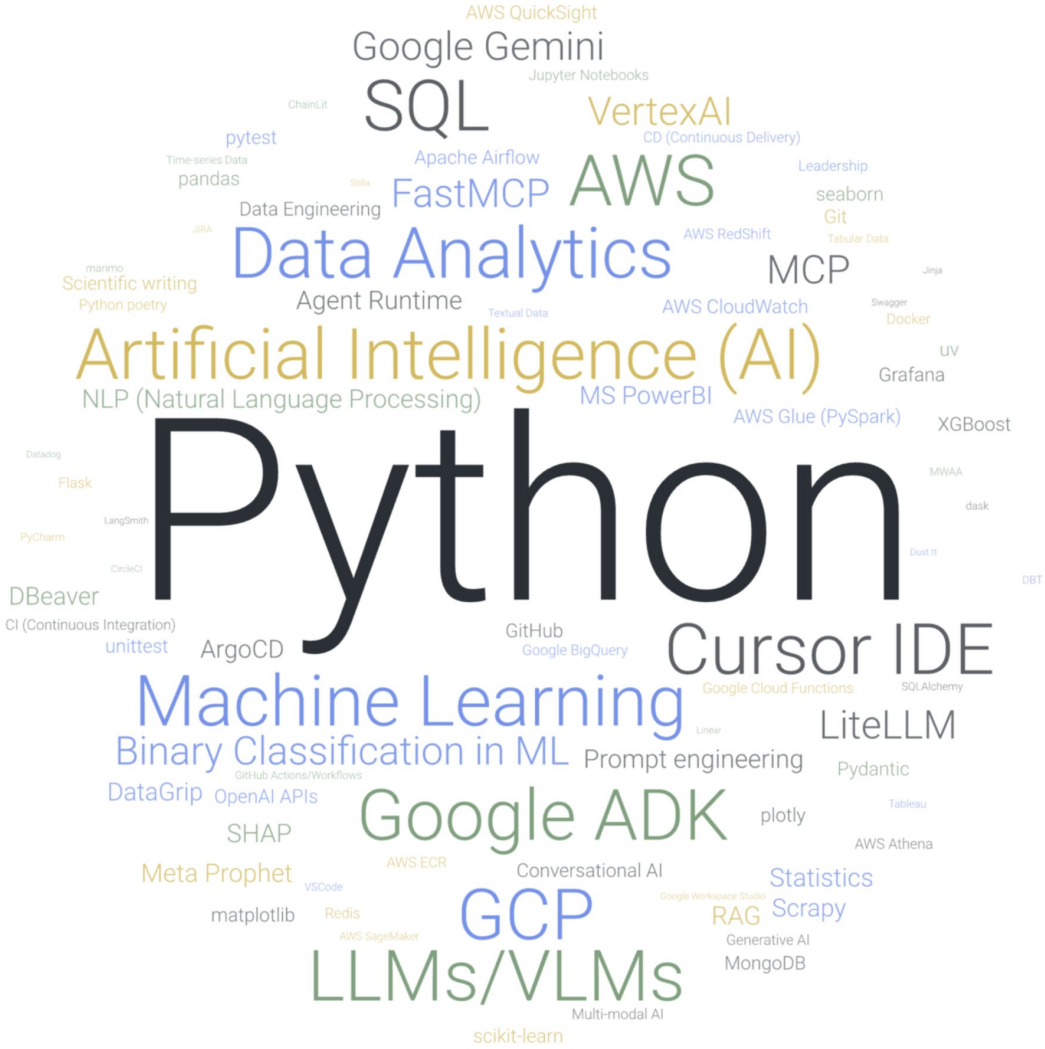
Technical Leadership

- Led applied AI, ML and data science delivery across fintech, payments, manufacturing and private equity, from discovery to shipped products.
- Turned ambiguous business problems into scoped technical plans, prototypes and shipped product capabilities.
- Balanced technical quality, delivery speed and business value across startup, scale-up and enterprise environments.
- Worked in successfully exited startups and scale-ups with a proven track record of business impact.

People Leadership

- Engineering management experience spanning 2+ years, including hiring, compensation, promotions and underperformance management.
- Led an AI/ML team of 4, hiring, onboarding and coaching senior, mid-level and junior data scientists.
- Worked closely with founders, senior leaders and customers, including CEO, CTO, Chief of Staff, CPO and VP-level stakeholders.
- Built ways of working that improved execution, ownership and growth, setting teams up for success.

Technologies



Contact

Vanja Smailovic

Stockholm, Sweden

vanja.smailovic@gmail.com

www.vanjasmailovic.com

